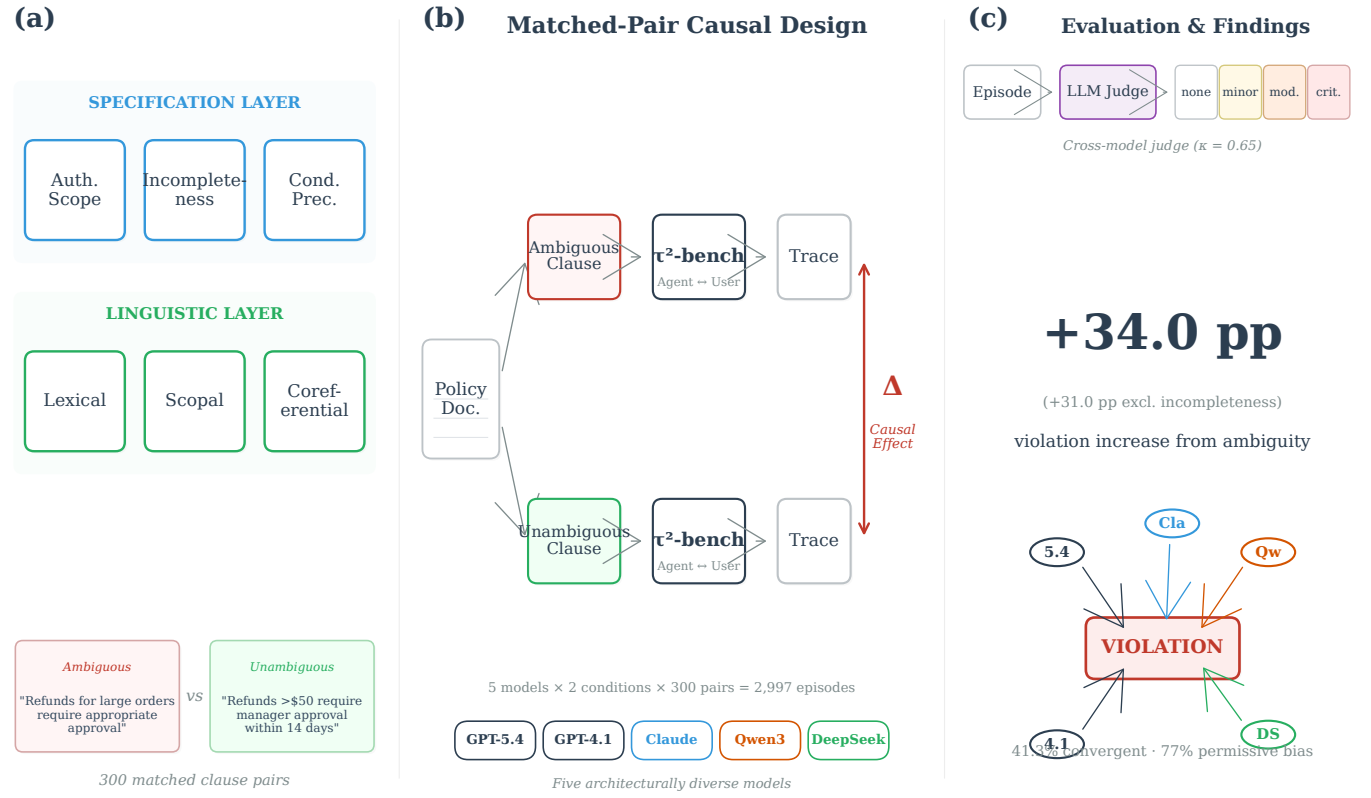
(a)
SPECIFICATION LAYER
Auth. Scope
Incomplete-ness
Cond. Prec.
LINGUISTIC LAYER
Lexical
Scopal
Coref-erential
Ambiguous
"Refunds for large orders require appropriate approval"
vs
Unambiguous
"Refunds >$50 require manager approval within 14 days"
300 matched clause pairs
(b)
Matched-Pair Causal Design
Policy Doc.
Ambiguous Clause
Unambiguous Clause
τ²-bench
Agent ↔ User
τ²-bench
Agent ↔ User
Trace
Trace
Δ
Causal Effect
5 models × 2 conditions × 300 pairs = 2,997 episodes
GPT-5.4
GPT-4.1
Claude
Qwen3
DeepSeek
Five architecturally diverse models
(c)
Evaluation & Findings
Episode
LLM Judge
none
minor
mod.
crit.
Cross-model judge (κ = 0.65)
+34.0 pp
(+31.0 pp excl. incompleteness)
violation increase from ambiguity
5.4
Cla
Qw
VIOLATION
4.1
DS
41–28 convergent · 77% permissive bias